

Identifying Post-COVID Risk Factors with Model-Agnostic Feature Importance

using the `xplainfi` R package with the German national cohort (NAKO)

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A matter of interpretation



Post-COVID condition (PCC)

Not to be confused with “long COVID”



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- Clinically heterogenous (symptoms & severity)

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Federal Ministry of
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What it can and can't tell us about PCC



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- “Importance” itself is not a single quantity → *method choice* defines the answer

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Permutation feature importance (**PFI**)

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 - $\rightarrow$ handles *mixed* data, *missing values*, computationally more expensive

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 - **Any method:** Repeat across resampling (e.g. bootstrap, subsampling, CV)
 - Captures variability from data sampling and learner stochasticity

Analysis dataset: NAKO

German National Cohort



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- $N \approx 66.000$ participants reporting ≥ 1 SARS-CoV-2 infection
- $\approx 24\%$ classified as “any PCC” (“Severe PCC” much rarer $\rightarrow$ not shown today)

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- ≈ 50 features used

Analysis: 'Any PCC' prediction

Preliminary, exploratory



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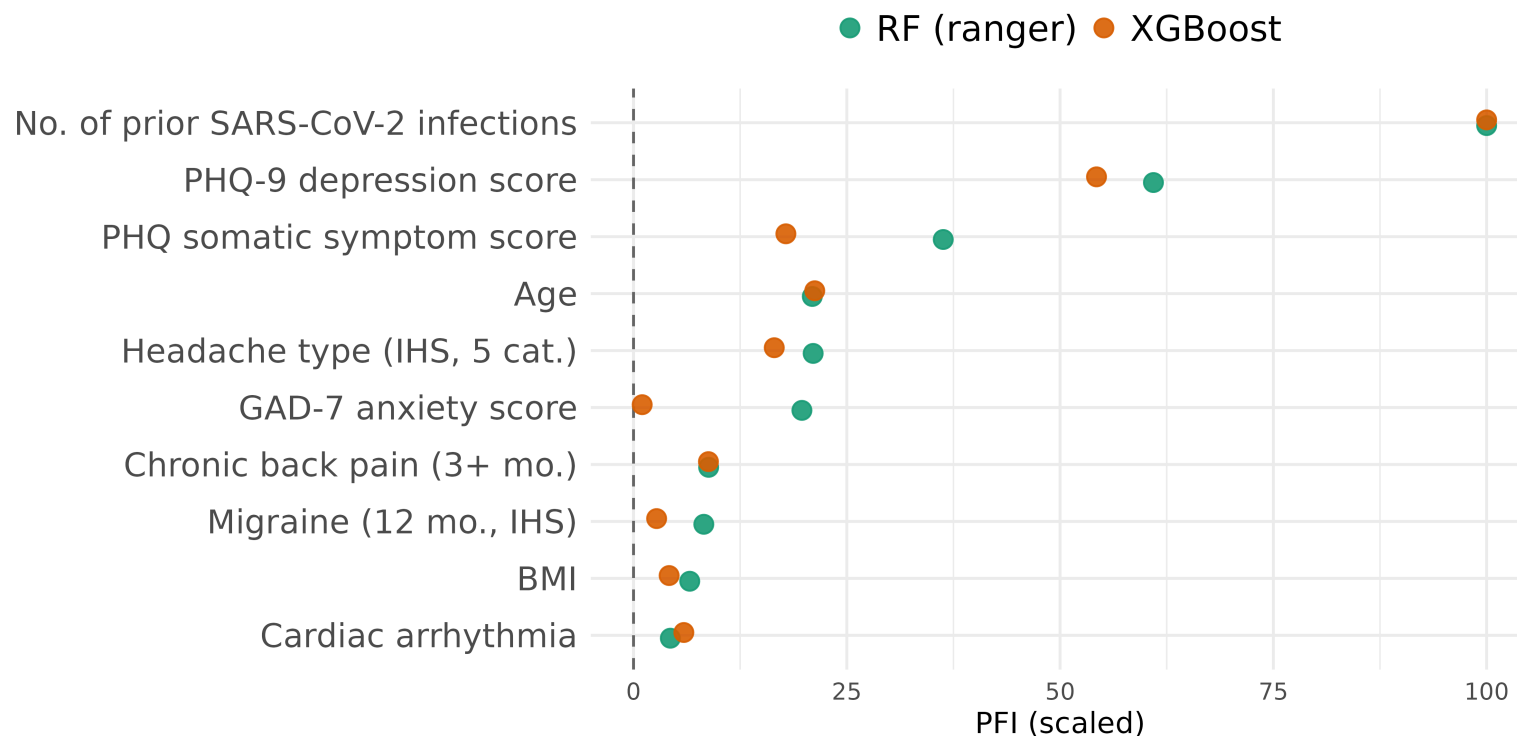


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- **PFI**, **CFI** (+ARF), **LOCO** computed on *test set* (model importance)

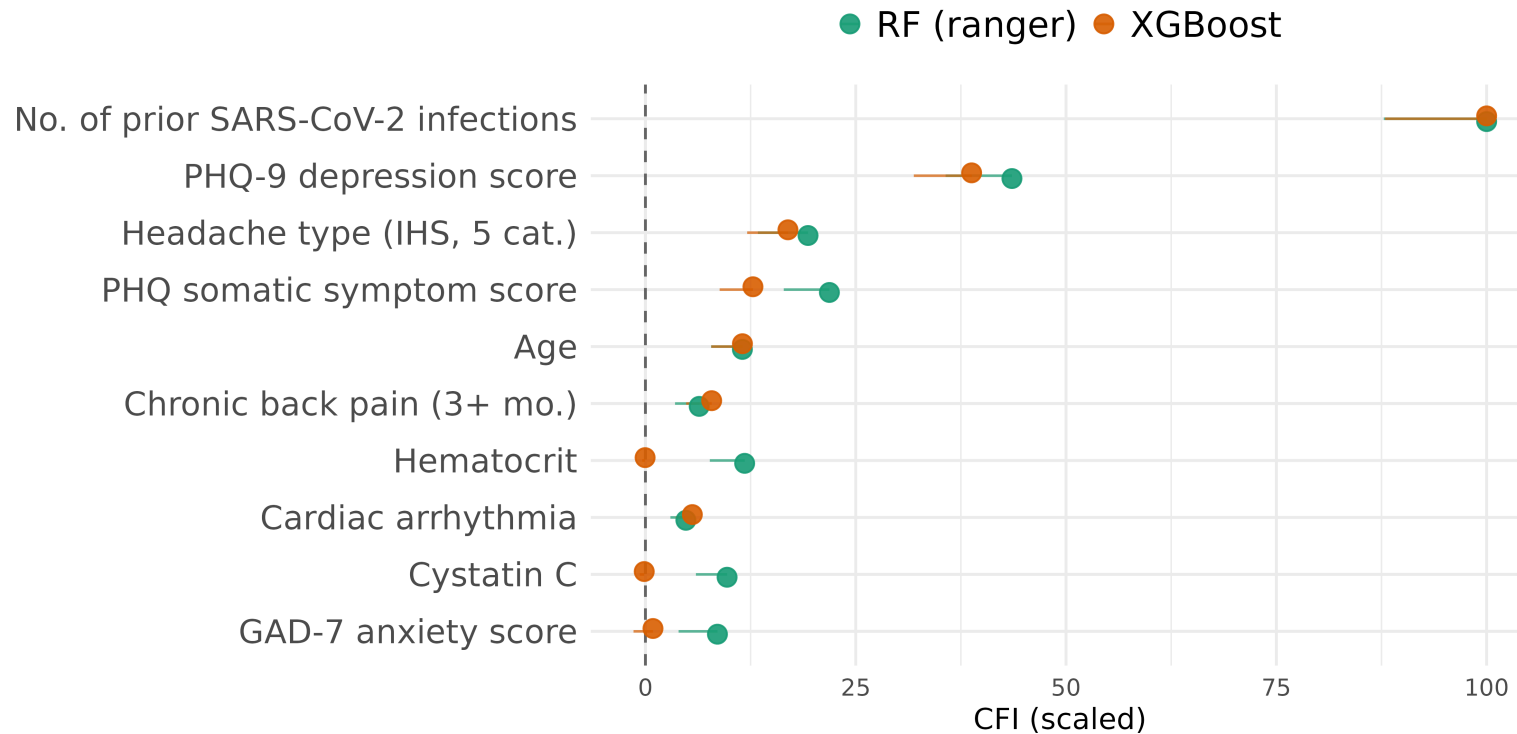
Results: Permutation Feature Importance (PFI)

Top 10 features (between RF + XGBoost)



Results: Conditional Feature Importance (CFI)

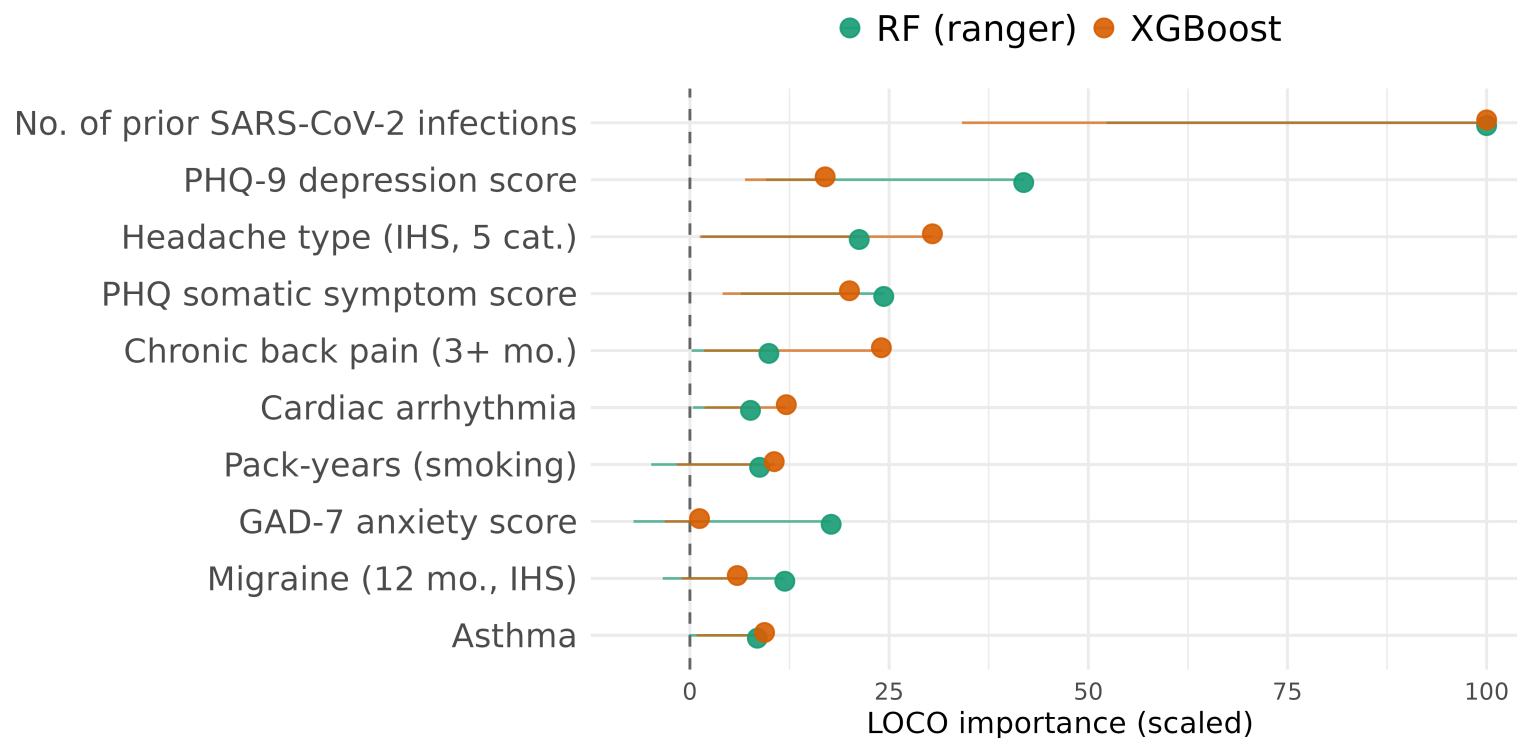
Top 10 features, one-sided 95% CIs



Results: Leave-one-covariate-out (LOCO)

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Thank you for your attention!



Contact

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